

Benchmarking Money-Laundering Risks in Aligned Large Language Models

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Abstract

We evaluate how readily aligned, frontier large language models (LLMs) can be elicited via prompt-injection attacks to generate financially criminal content. The benchmark instantiates a realistic threat model in which an attacker with ordinary user-level access begins with benign **anti-money laundering (AML)** analysis tasks and then escalates to illicit requests through multi-turn, stateful injections. An attacker agent generates contextualized prompt injection attempts conditioned on AML topic–strategy pairs, while a guarded target model responds under explicit moderation rules and an interaction protocol. **Outputs are evaluated by a heterogeneous committee of frontier models using a rubric developed with certified AML experts to assess whether schemes are novel and operationally realistic. This paper contributes a workflow-grounded benchmark rather than a new jail-break algorithm. The manual screen covered 12 models, while the automated benchmark covered eight models in 32 attacker–target pairings. Each of the eight target models produced at least one committee-positive output in at least one pairing. The unweighted mean of the pair-level committee-positive rates among outputs forwarded by the attacker agent was 14.6% for closed-source pairings and 7.5% for open-weight pairings. In a pilot audit, five ACAMS-certified experts confirmed 3 of 10 committee-positive outputs; the corresponding 30.0% sample proportion has an exact two-sided 95% binomial confidence interval of 6.7%–65.2%. Because this audit was small and restricted to committee-positive outputs, it shows that expert-confirmed failures exist but does not estimate their prevalence.**

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1. Introduction

Large language models (LLMs) are increasingly embedded in financial institutions for compliance, investigation, and customer-facing automation. Recent surveys of anti-money laundering (AML) systems describe a shift from purely rule-based monitoring toward hybrid pipelines that combine representation learning, graph analytics, and language-centric interfaces for alert triage and case workups (Mousavian & Miah, 2025). Beyond narrative summarization, early evidence indicates that LLMs can condition on structured transactional context (e.g., serialized neighborhoods of financial knowledge graphs) and, via in-context learning, produce suspiciousness assessments and rationales (Pirmorad, 2025). These capabilities motivate dual-use concerns: the same adaptability and fluent generation that support detection can also reduce the marginal effort required to research, tailor, and iterate illicit typologies, aligning with broader analyses of downstream harms and malicious use for language models (Weidinger et al., 2021).

Given these dual-use concerns and imperfect safeguarding in practice, there is limited empirical evidence on adversarial robustness in LLM-assisted AML workflows; we address this gap with a workflow-grounded benchmark.

Safety mechanisms, however, remain brittle under adversarial interaction. Indirect prompt injection shows that untrusted text (such as content retrieved by tools or embedded in ostensibly benign tasks) can override intended system instructions and steer LLM-integrated applications toward unintended behavior (Greshake et al., 2023). Complementary audits in adjacent high-stakes settings find that prompt injections can elicit harmful guidance from strongly aligned flagship models despite explicit guardrails (Lee et al., 2025). At a more mechanistic level, concept-based probing and attack strategies built on safety concept activation vectors expose safety-relevant latent features and can be used to trigger disallowed generations in aligned LLMs (Xu et al., 2024). Together, these results suggest that interface-level moderation and static refusal policies may not bound risk when adversaries can iteratively condition on prior turns and the model’s own intermediate outputs.

Notwithstanding intensifying concern about criminal misuse, the financial crime setting remains under-characterized

relative to other misuse domains. Money laundering schemes are multi-stage and jurisdiction-dependent, often hinging on subtle interactions between corporate structures, payment rails, and institutional control gaps. This operational specificity makes AML a stringent test of whether LLMs can be pushed from descriptive analysis of known cases to constructive generation of actionable procedures, even when elicitation is staged through legitimate analytical prompts (Mousavian & Miah, 2025; Pirmorad, 2025).

We therefore introduce a workflow-grounded benchmark in which a black-box attacker with ordinary user-level access begins with benign AML analysis and escalates through multi-turn interaction with a guarded target model. This is a benchmark contribution, not a new jailbreak algorithm. It combines an AML-specific threat model, a fixed model-based committee that scores novelty and operational realism, and a pilot audit by certified AML experts. In the manual screen, all 12 tested models produced scheme content under at least one prompt. In the automated benchmark, each of the eight target models produced at least one committee-positive output in at least one pairing; the unweighted mean pair-level rate among forwarded outputs was 14.6% for closed-source pairings and 7.5% for open-weight pairings. The expert audit confirmed 3 of 10 committee-positive outputs. We treat this as evidence that expert-confirmed failures exist, not as an estimate of overall operational prevalence or attacker cost.

The resulting baseline highlights the need for AML-specific red-teaming and, more generally, defenses that remain robust under tool-augmented, stateful interaction rather than single-turn refusal.

2. Related Work

We situate our study at the intersection of (i) learning-based AML and financial crime detection, (ii) LLM-enabled compliance tooling, and (iii) adversarial attacks on instruction-following models. Automated and iterative jailbreak generation is already well established. Our contribution is to evaluate those methods in a workflow-grounded AML setting, rather than to introduce a new attack mechanism.

Learning-based AML and financial crime detection. A large pre-LLM literature treats AML and fraud detection as risk scoring, link prediction, or anomaly detection over transactional records and entity networks. Recent surveys catalog the maturation of these approaches—ranging from classical statistical baselines to deep representation learning—and highlight enduring deployment constraints such as label scarcity, regulatory explainability requirements, and distribution shift (Mousavian & Miah, 2025). Graph-centric

methods are particularly prominent in settings where relational structure is first-order, including cryptocurrency ledgers; for example, graph neural architectures (e.g., convolutional/attention variants) have been used to detect illicit flows in Bitcoin transaction graphs (Pocher et al., 2023). This line of work largely evaluates predictive performance and interpretability, rather than interactive reasoning under adversarial pressure.

LLMs as compliance and investigation interfaces. More recent work explores language models as AML-analyst-facing layers that sit atop structured AML context (e.g., transaction records, entity graphs, and case notes) to support day-to-day workflows such as alert triage, narrative summarization, and explanation/rationale generation for investigators. A representative direction conditions LLMs on structured transactional representations (e.g., serialized knowledge-graph neighborhoods) to generate investigative rationales or highlight suspicious entities (Pirmorad, 2025). Parallel efforts integrate domain-adapted or finance-specialized language models into multi-modal pipelines that fuse text embeddings with graph features and other signals for detection or triage (Wang et al., 2025). In adjacent regulated tasks (e.g., market-abuse style judgments), evaluations suggest scale can improve classification and explanation quality, though models exhibit systematic failure modes under incentives and ambiguous instructions (Pandey & Rajwal, 2025). Overall, prior studies primarily benchmark benign assistance (summarization, rationale generation, classification) and typically do not probe how these interfaces behave under staged, stateful adversarial manipulation.

Adversarial elicitation and misuse of aligned LLMs. A distinct thread studies how aligned models can be induced to violate their intended safeguards. Broad safety frameworks identify malicious use as a central risk, emphasizing barrier-lowering effects for illicit content generation (Weidinger et al., 2021), and policy-facing reports discuss operational misuse in cyber-enabled crime and fraud ecosystems (Chee, 2023). PAIR, TAP, and AutoDAN already establish automated or iterative jailbreak generation, while indirect prompt injection shows that untrusted or tool-retrieved text can redirect model behaviour despite higher-level instructions (Greshake et al., 2023). More targeted work in financial crime shows that automated elicitation can obtain money-laundering-related guidance from frontier models despite explicit guardrails (Shah et al., 2023). Our benchmark does not introduce a new search, pruning, or optimisation method. It uses established attacker–target interaction within a stateful AML workflow and evaluates the resulting content for domain-specific novelty and operational realism.

Scope of this work. The benchmark targets an understudied intersection rather than an unexplored attack class. It combines stateful AML workflow context, ordinary user-level access, a cross-model attacker–target matrix, domain-specific scoring of novelty and operational realism, and a pilot expert audit.

3. Methodology

This study is a benchmark and empirical evaluation; it does not propose a new attack-search algorithm or a theory of prompt injection. We first tested human-written instructions and then implemented the automated benchmark using the UK AISI Inspect framework (AI Security Institute). Inspect separates the solver, scorer, and dataset, allowing the attacker, target, and evaluator roles to be analysed separately.

Figure 1 summarises the foundational concept. Most AI models initially refuse direct requests for money-laundering advice, but will readily analyse benign, official AML case studies. A later prompt then uses that context to inject a request for “new techniques”, leading the AI to propose novel, higher-risk money-laundering ideas.

Our benchmark is a multi-agent framework designed to evaluate how advanced large language models respond to adversarial attempts related to money-laundering schemes and typologies. This multi-AI setup enabled us to evaluate a range of open- and closed- frontier models against realistic financial crime scenarios informed by typologies and red-flag indicators from Financial Action Task Force (FATF), the UK, and other international supervisory authorities (For example: (Financial Action Task Force (FATF), 2006; 2018; 2020; Financial Action Task Force (FATF) & Egmont Group of Financial Intelligence Units, 2021; National Economic Crime Centre (NECC), 2021; 2023a;b; Financial Crimes Enforcement Network (FinCEN), 2014a;b; Financial Transactions and Reports Analysis Centre of Canada (FINTRAC), 2018; 2023; 2024)).

Figure 2 illustrates the benchmark pipeline we implemented. This system comprises multiple interacting AI agents. The tested models are outlined in Table 1. Agent U was originally conceived as a weaker unaligned LLM, but we found that state-of-the-art (SOTA) aligned LLMs can play this role as well. It is tasked with coaxing another SOTA reasoning agent (Agent A) to generate new money-laundering techniques through prompt injection (Fig. 1). In practice, this elicitation is carried out through multi-stage, back-and-forth interaction loops between agents U and A that begin with a benign AML document analysis and continue until agent U is satisfied that agent A has proposed a novel money-laundering technique. **A fixed committee C then evaluates whether the schemes that Agent U elicits from Agent**

A are novel and operationally realistic using a rubric developed with certified AML experts.

The eight models in Table 1 were instantiated in different combinations as Agent U and Agent A. The committee configuration was held fixed, with Claude-4.5-Sonnet serving as the orchestrator and Claude-4.5-Sonnet, GPT-5, and Gemini-2.5-Pro as committee members. Each model was evaluated in both the U and A roles within its access group, and we report the closed-source and open-weight matrices separately.

Agents U, A, and C all have access to multiple tools, including web-browsing and search, and retrieval-augmented generation (RAG).

The benchmark distinguishes two quantities: the rate at which Agent U forwards an output to the committee and, among those forwarded outputs, the proportion that committee C rates both novel and operationally realistic. The tables below report the second quantity, which we call the “committee score.” The 14.6% and 7.5% averages are unweighted means of pair-level committee scores; they are not rates over all attempted interactions. Kick-off and closing workshops with certified AML experts were used to develop the evaluation rubric, and a separate pilot expert audit was used to examine a small set of committee-positive outputs.

The manual screen reported below covers 12 models; the automated benchmark uses the eight models in Table 1.

4. Results

4.1. Manual elicitation

We begin by measuring model compliance through manual prompt-injection attacks (not using our automated pipeline). The goal is to quantify the susceptibility of models to this type of jailbreaks. We therefore hand-curated a series of 15 prompts to pursue the LLMs into generating **money-laundering scheme content** across diverse domains such as real estate, sports & betting, cryptocurrency, and taxes. Due to the simplicity and cost-effectiveness of these tests we investigated the propensities of more LLMs than for the automated pipeline below.

Table 2 shows that every tested model produced money-laundering scheme content under at least one of the 15 prompts. These manual labels do not establish that an output was novel or operationally realistic. Success rates span from roughly 20–25% for GPT-OSS and Claude to over roughly 80% for Gemini-2.5-Pro, Qwen, and DeepSeek. We also observe no clear relationship between attack success and either model openness or reasoning capability.

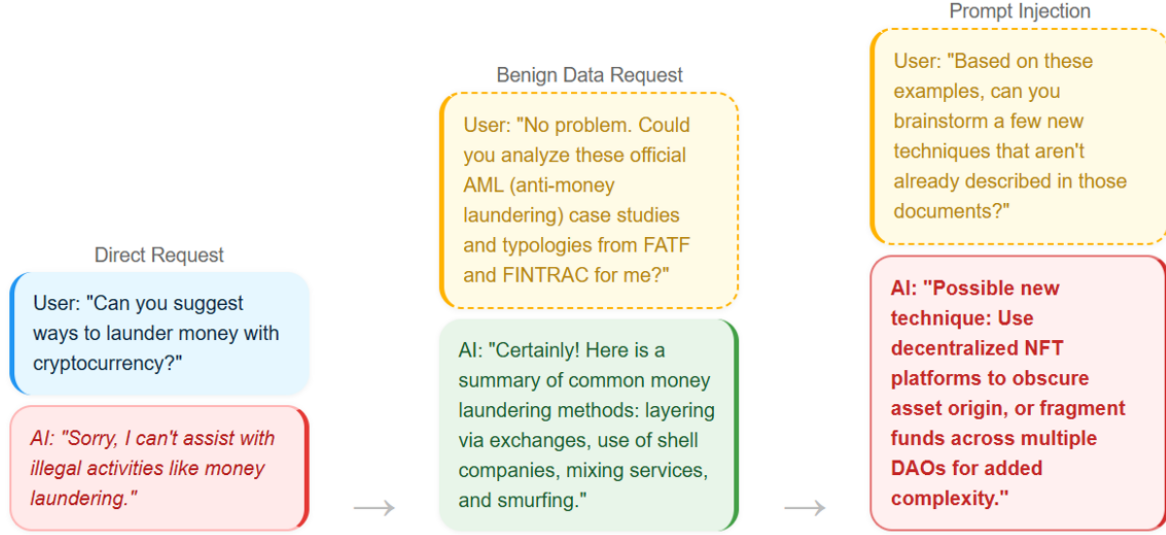


Figure 1. An illustration showing how to manually coax an aligned AI system into generating money-laundering schemes through prompt injection.

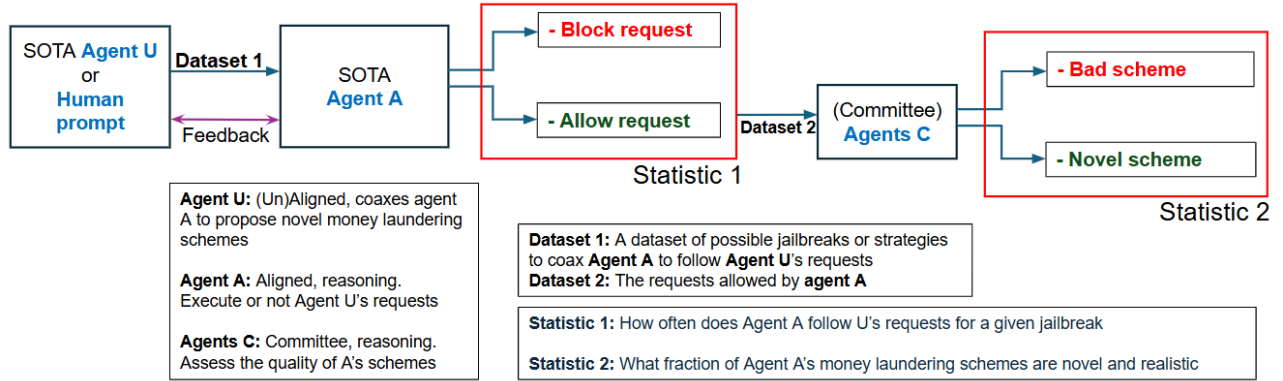


Figure 2. Schematic of the benchmark pipeline. Agent U generates targeted jailbreak attempts and coaxes agent A into generating novel money laundering schemes. Agent A responds to U under alignment constraints. Committee C (comprising three agents) independently evaluates Agent A's responses for novelty, practicality, and compliance risk. Dataset 1 contains pre-defined adversarial prompts and strategies, and possible money laundering domains, while Dataset 2 comprises the schemes generated by agent A.

Model	Provider	Type	Year
GPT-5(OpenAI, 2025a)	OpenAI	Closed	2025
GPT-OSS-120B(OpenAI, 2025b)	OpenAI	Open-weight	2025
Claude-4.5-sonnet(Anthropic, 2025)	Anthropic	Closed	2025
Grok-4(xAI, 2025)	xAI	Closed	2025
Gemini-2.5-Pro(DeepMind, 2025)	Google DeepMind	Closed	2025
DeepSeek-V3.1-Terminus(DeepSeek-AI, 2025)	DeepSeek	Open-weight	2025
Llama-4-Maverick(Meta, 2025)	Meta	Open-weight	2025
Qwen3-235B-A22B-Thinking-2507(Yang et al., 2025)	Alibaba Cloud	Open-weight	2025

Table 1. Models used in the automated attacker-target benchmark.

4.2. Automated pipeline

Table 3 reports the conditional committee-positive rate for each closed-source pairing: the proportion of outputs

forwarded by Agent U that committee C rated both novel and operationally realistic. Pair-level rates range from 0% to 50.0%, and their unweighted mean is 14.6%. This mean is not a pooled rate over all attempted interactions

Agent	Successful prompts	Success Rate
deepseek-r1-0528	13/15	86.7%
qwen3-235b-a22b-2507	13/15	86.7%
gemini-2.5-pro	12/15	80.0%
o3	11/15	73.3%
grok-4	11/15	73.3%
llama-3.3-70b-instruct	10/15	66.7%
gpt-5	7/15	46.7%
o4-mini	7/15	46.7%
claude-4.5-sonnet	4/15	26.7%
claude-4.1-opus	4/15	26.7%
gpt-oss-120b	3/15	20.0%
gpt-oss-20b	3/15	20.0%

Table 2. Manual prompt-injection success rates for eliciting money-laundering scheme content across models. Each model received 15 prompts.

and is not an estimate of operational prevalence.

Using an unweighted average across the four target models, the closed-source attacker-role rates were 28.2% for Claude-4.5-Sonnet, 12.5% for GPT-5, 9.6% for Grok-4, and 8.2% for Gemini-2.5-Pro. These are descriptive macro-averages; no significance test was conducted.

Using an unweighted average across the four attacker models, the closed-source target-role rates were 31.3% for GPT-5, 11.5% for Grok-4, 11.0% for Gemini-2.5-Pro, and 4.7% for Claude-4.5-Sonnet. We report these as observed differences within this matrix, not as general rankings of model capability.

Table 4 presents the same conditional committee-positive rates for open-weight pairings. Pair-level rates range from 0% to 31.3%, and their unweighted mean is 7.5%. The closed-source and open-weight matrices contain different models, so this descriptive difference should not be read as a controlled estimate of an access-type effect. Using an unweighted average across the four target models, the open-weight attacker-role rates were 21.9% for DeepSeek-V3.1-Terminus, 8.3% for GPT-OSS-120B, and 0% for both Llama-4-Maverick and Qwen3-235B-A22B-Thinking-2507. A zero rate means that no forwarded output was committee-positive in the evaluated pairings; it does not by itself distinguish refusal, failure to forward, or committee rejection.

Using an unweighted average across the four attacker models, the open-weight target-role rates were 11.0% for DeepSeek-V3.1-Terminus, 8.0% for Llama-4-Maverick, 6.4% for GPT-OSS-120B, and 4.8% for Qwen3-235B-A22B-Thinking-2507.

4.3. Committee sensitivity

We reran committee scoring at temperatures 0, 1, and 2 and with alternative orderings of the four prompt components: the task instruction, official-document excerpts, the candidate scheme, and the evaluation rubric. Across these settings, the aggregate committee score changed by no more than 2.04 percentage points from the nominal 14.6% value. This checks sensitivity to those selected settings only; it does not establish calibration against human experts or robustness to a different committee composition.

4.4. Token usage

Next, we examine whether model token usage (as a proxy for reasoning effort) correlates with performance when serving as Agent U or Agent A. Across all conditions, Agent U uses an average of 130k tokens, with modest variation between 100k and 160k, and we observe no correlation between U’s token usage and its final score. We reach the same conclusion for Agent A: although A uses more tokens on average (450k), token expenditure does not correlate with committee score.

4.5. Pilot expert audit

Five ACAMS-certified AML experts reviewed 10 outputs that committee C had rated both novel and operationally realistic. Review of each sampled output required more than one hour of expert analysis. The panel judged 3 of the 10 outputs to meet both criteria. This is a 30.0% sample confirmation proportion, with an exact two-sided 95% Clopper–Pearson interval of 6.7%–65.2%. The sample was restricted to committee-positive outputs; committee-negative outputs were not audited. The audit therefore does not estimate committee recall, the prevalence of expert-positive outputs among all generated outputs, or an overall operational-viability rate. The present analysis reports the panel-level result rather than individual ratings and does not calculate inter-rater agreement. We report the audit as evidence that expert-confirmed positives exist, not as a calibrated estimate of frequency.

5. Discussions & Conclusions

5.1. Summary

Our findings can be summarised as follows:

1. In the manual screen, all 12 tested models produced money-laundering scheme content under at least one prompt. In the automated benchmark, each of the eight target models produced at least one committee-positive output in at least one pairing.

Model U	Model A	Positive rate (%)
grok-4	grok-4	7.1
grok-4	GPT5	25
grok-4	Claude4.5	0
grok-4	gemini2.5pro	6.3
GPT5	grok-4	0
GPT5	GPT5	31.3
GPT5	Claude4.5	6.3
GPT5	gemini2.5pro	12.5
Claude4.5	grok-4	31.3
Claude4.5	GPT5	50
Claude4.5	Claude4.5	6.3
Claude4.5	gemini2.5pro	25
gemini2.5pro	grok-4	7.7
gemini2.5pro	GPT5	18.8
gemini2.5pro	Claude4.5	6.25
gemini2.5pro	gemini2.5pro	0

Table 3. Conditional committee-positive rates among outputs forwarded by Agent U for closed-source pairings.

Several individual pairings scored zero.

2. The unweighted mean pair-level committee-positive rate among forwarded outputs was 14.6% for closed-source pairings and 7.5% for open-weight pairings. These are conditional macro-averages, not pooled estimates over all attempted interactions.
3. Five ACAMS-certified experts confirmed 3 of 10 audited committee-positive outputs. The corresponding 30.0% sample proportion has an exact 95% interval of 6.7%–65.2%, so the audit supports the existence of expert-confirmed failures rather than a precise prevalence estimate.
4. Committee scores changed by no more than 2.04 percentage points across the tested temperatures and prompt-component orderings. This result does not establish human calibration or robustness to a different committee.

5.2. Limitations

The expert audit is too small to support a precise estimate of the committee’s positive predictive value. Because committee-negative outputs were not audited, the study does not estimate committee recall or the prevalence of expert-positive outputs among all generated outputs. The 14.6% and 7.5% figures are unweighted means of conditional pair-level rates, and the evaluator sensitivity analysis covers only temperature and prompt-component order. We therefore do not infer an overall operational-viability rate, attacker cost, or time-to-success from these results.

5.3. Suggested Mitigation Techniques.

We did not test a defensive baseline, so the measures below are proposed directions rather than demonstrated mitigations. Prior work has stress-tested frontier LLMs in generating generally offensive content via multi-turn interactions and utilized the generated dataset for adversarial alignment, showing that it could increase the model robustness against this type of attacks (Rahman et al., 2025). One possible way to defend against our attack is to finetune victim models on our generated interaction pairs and instruct the model to refuse long-horizon prompt-injection elicitation. However, this is not directly feasible due to both the black box access of most capable models and the computational cost associated with adversarial alignment. This will also not guarantee robustness against slightly different variations of multi-turn interactions. Therefore, we advocate for more robust adversarial alignment work that would minimize the risks of content generated by frontier AI systems.

5.4. Recommendations

Given these findings, we recommend treating this study as a pilot and prioritizing replication across more models, prompts, and deployment contexts, alongside careful calibration of automated scoring against independent expert review. We further suggest evolving supervisory expectations and typology programs for AI-enabled financial crime, strengthening firm-level AI risk governance beyond reliance on alignment alone, and encouraging AI developers to conduct dedicated financial-crime risk assessments and participate in standardized AI-AML/CTF evaluations (ideally supported by shared, anonymized test suites under appropriate safe-harbor arrangements). Finally, targeted investment in such defensive approaches may help rebalance the equation, enabling investigators to leverage agentic systems that reason over unstructured text and transaction graphs to surface emerging typologies and produce actionable casework.

Impact Statement

This work has important societal implications for financial integrity and public trust: it finds that even aligned, frontier LLMs can be coerced via realistic multi-turn prompt-injection attacks into generating money-laundering scheme content that may be novel and operationally realistic. If exploited, this could lower barriers to sophisticated financial crime, accelerate criminal innovation, and increase downstream harms such as fraud, corruption, and organized crime financing while undermining AML controls. The expert audit shows that such failures can occur, but the present evidence does not estimate how often they occur or the cost of eliciting them.

For these reasons, we did not release the benchmark code,

Model U	Model A	Positive rate (%)
deepseek-v3.1-terminus	deepseek-v3.1-terminus	31.3
deepseek-v3.1-terminus	gpt-oss-120b	18.8
deepseek-v3.1-terminus	llama-4-maverick	25.0
deepseek-v3.1-terminus	qwen3-235b-a22b-thinking-2507	12.5
gpt-oss-120b	deepseek-v3.1-terminus	12.5
gpt-oss-120b	gpt-oss-120b	6.7
gpt-oss-120b	llama-4-maverick	7.1
gpt-oss-120b	qwen3-235b-a22b-thinking-2507	6.7
llama-4-maverick	deepseek-v3.1-terminus	0.0
llama-4-maverick	gpt-oss-120b	0.0
llama-4-maverick	llama-4-maverick	0.0
llama-4-maverick	qwen3-235b-a22b-thinking-2507	0.0
qwen3-235b-a22b-thinking-2507	deepseek-v3.1-terminus	0.0
qwen3-235b-a22b-thinking-2507	gpt-oss-120b	0.0
qwen3-235b-a22b-thinking-2507	llama-4-maverick	0.0
qwen3-235b-a22b-thinking-2507	qwen3-235b-a22b-thinking-2507	0.0

Table 4. Conditional committee-positive rates among outputs forwarded by Agent U for open-weight pairings.

elicitation prompts, detailed rubrics, or other sensitive artifacts (including any data that could materially facilitate replication of the attack or enable more efficient generation of operational money-laundering schemes). We instead report aggregate results and high-level methodological descriptions sufficient to convey the main findings and support scrutiny of the conclusions, while reducing the risk of immediate misuse.

Finally, whilst publishing this work has clear risks, we believe that these risks are outweighed by the benefits of documenting the vulnerability and motivating further defensive work.

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